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Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHONOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

Sr.No. 8522

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:401 Course Section:1	Seat No: 2022600038

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Name of the candidate. Parnika Das

Candidate's UID number: 2 0 2 2 6 0 0 0 3 8

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH Mid Semester Examination
(For example FY MCA/FY BTECH)

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE AIML

DAY AND DATE: 08th October 2025, Wednesday

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

8255

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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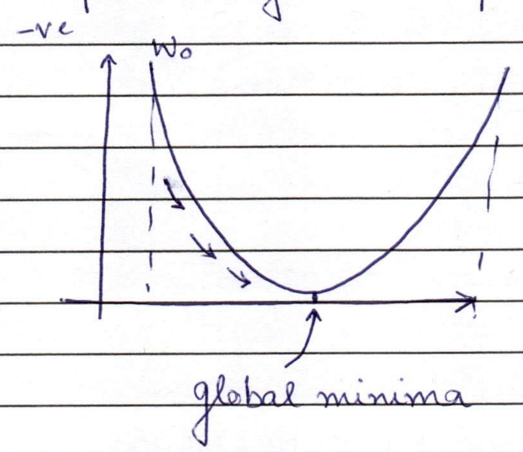
Q1.

a. $f(w) = \frac{1}{2} \| Xw - y \|^2$

As per the gradient update rule



To acheive the proper weighted value, we backpropagate to the value nearing to the global minima.

$$\theta^{k+1} = \theta^{k} - \eta \nabla f(\theta^k)$$

learning rate

$$\nabla f(\theta^k) = \frac{\partial L}{\partial f(w)}$$

$$f(w) = \frac{1}{2} \| Xw - y \|^2 = \frac{1}{2} \left[X^2 w^2 + y^2 - 2Xwy \right]$$

$$\frac{\partial f(w)}{\partial w} = \frac{1}{2} \left[2X^2 + y^2 - 2Xy \right] = X^2 + y^2/2 - Xy$$

$$\frac{\partial f(w)}{\partial y} = \frac{1}{2} \left[X^2 w^2 + 2y - 2Xw \right]$$

$$= \frac{X^2 w^2}{2} + y - Xw.$$

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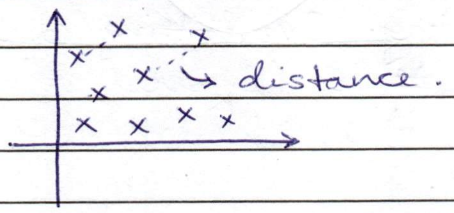
Q1.

b. As per the neural network function

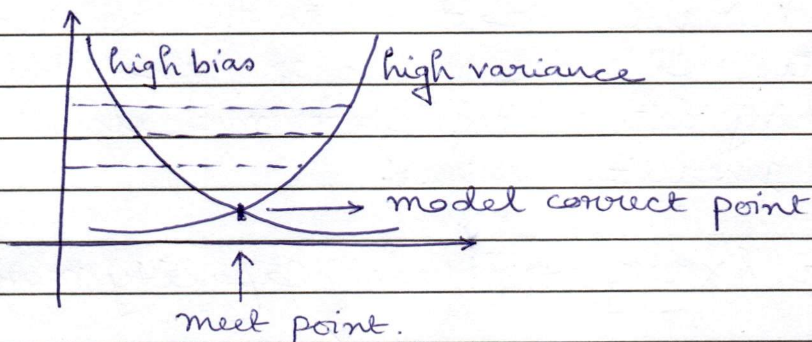
$\Rightarrow \sigma(w * x + b)$

where $w \rightarrow$ weight value.
$x \rightarrow$ input value.

So when the model is being trained, the model checks the distance between the given values.



Similar way, to predict the error value, it checks the error distance.



So to check the error, we also look for the value that was predicted & the actual value.

And because of this graph, The ~~error~~ Sumation of square value is taken.

$$\sum_{i=0}^{n} |y-\hat{y}|^2$$

where $y \Rightarrow$ predicted value.
$\hat{y} \Rightarrow$ actual value.

This error calculation helps us to find the loss value of the function.

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When the loss value is calculated. The value helps to predict how the model is being trained.

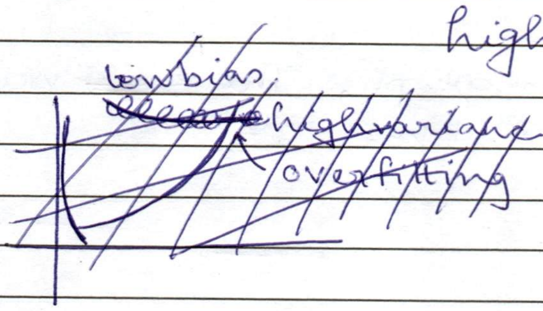
Lower the loss value, better the model would work for testing data.

But for very ~~loss~~ low loss value can create a concern. It is the marking of the model learning not only the trained value but also the extra unecessary datas & noise.

This leads to overfitting. Overffitting can also be predicted from the ~~M~~ SE (Squared error value) lower the value, it says how much of the model is overfitted.

Overfilting is lead by high varience. This also effects the generalization as the unecessary data and noise values are also considered as training data which leds to mis calculations of the generalized values.

Lower error ⇒ Model Capacity more high
↓
leads to Overfitting
↓
high varience.



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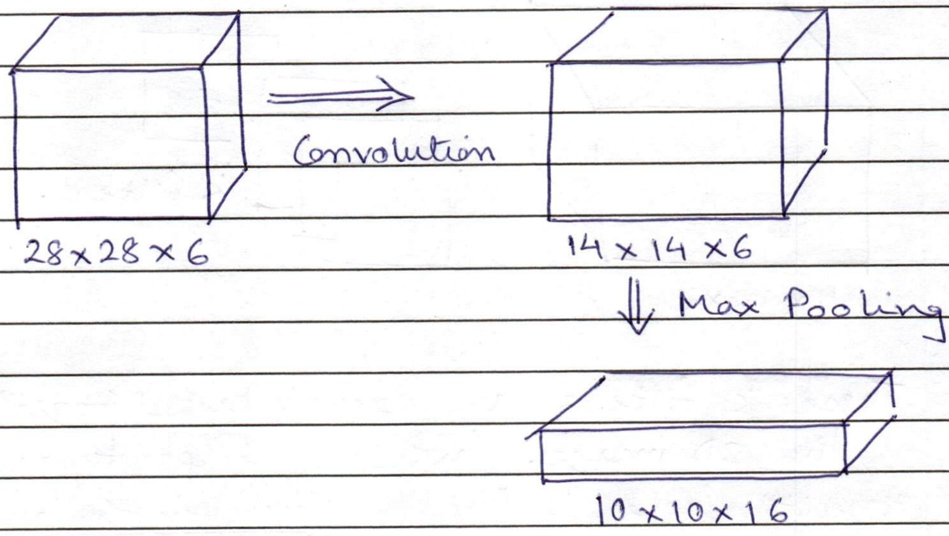
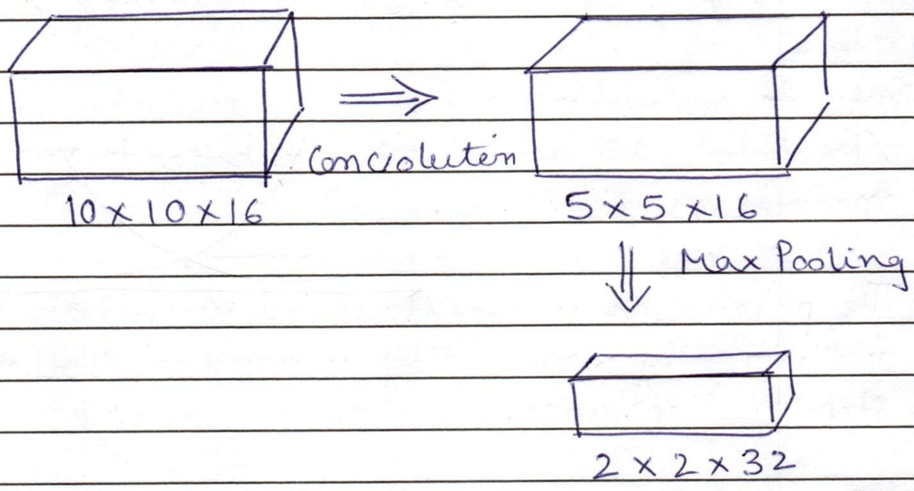
Q2.

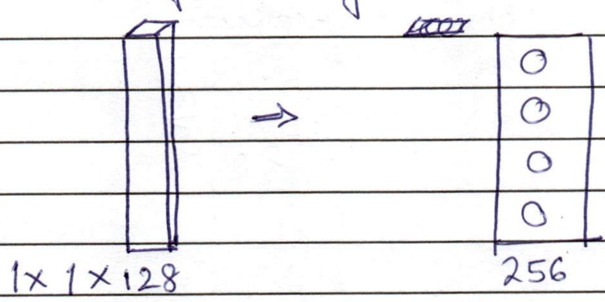
a. Standard 2D convolution



Again we will go with convolution



We will repeat the convolution & the max pooling step again & again till we reach the 1D convolution



We can do the basic convolution on the 1D convoluteon and we can find the values.

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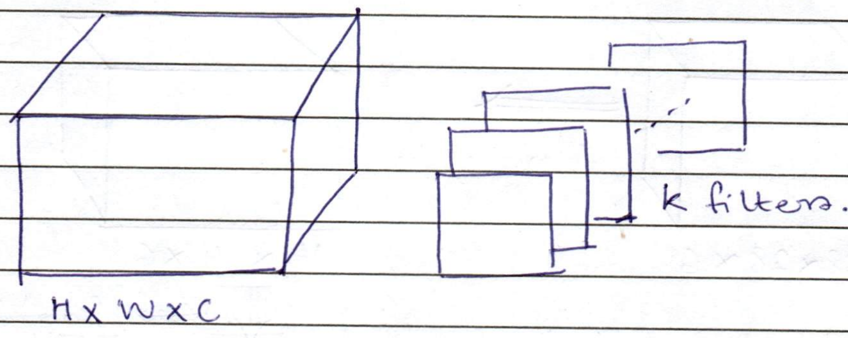
A depthwise Seperable Convolution.

input size H x W x C   K filters of F x F



These k filters are convolutated together over the 2D image, which helps to achive results faster. But the issue in the modern point is that this method can lead to multiple errors & issue arising with this. It can lead to wrong prediction. It is a fast & Sloopy method.

But to priortice accuracy and better result Standard 2D convolution is better to work in modern architecture.

The parameters in the standard method are handled individually and they worked on the model step to step wise.

There is no skip on the step that would lead to missing out few parameters and lead to the not so correct accurate values.

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Q2.

b. Over complete autoencoders: The autoencoders with the feed forward neural network tend to have more hidden layers ~~nodes~~ than the input values. This leads to more complexity & less accuracy of output results.

identity function ⇒ $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Now here if proper regularization is not done, the no. of hidden values will be unchanged. It will demand for more and more input values.

Here the input values being lesser than the hidden layer would be exhausted. So, the hidden values tends to start training on empty identity values.

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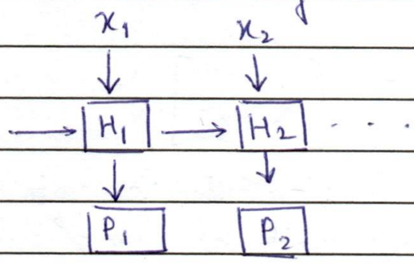
Q3.

a. $\dfrac{\partial L}{\partial h_{t-k}}$

In RNN (Recurrent Neural Network) it is a type of a neural network that also considers the value of the ~~next~~ previous sequencial value.



The basic ~~liear~~ linear equation states that.

$$h_t = f[W*(h_{t-1}) + W*x_t + b]$$

$$h_t = h_{t-1} + \frac{\partial L}{\partial h_{t-1}} \Big\} \text{ loss derivative function}$$

But as we go further, the derivative loss value increases. The increase in loss value is not considered good.

The LSTM ⇒ Long Short Term Memory helps to maintain the loss derivate value. It introduces the forget gate function which helps in maintain the loss value.

$$f_t = \sigma[W_f * (h_{t-1}, x_t) + b]$$

this comparision helps to maintain the values as the one with the lesser loss derivate value will be considered to proceed ahead with.

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Q3. b.

GRU → Grated Reccurrent Unit.

It has ~~to~~ two gates Update & Reset. The Update gate updates the value of the model while the reset gate works when we see an ~~1st~~ issue or wrong direction prediction, it has the capability to reset to the previous value.

$z_t = \sigma(w^*_{z_t}(h_{t-1}, x) + b)$ where b is the bias.

$r_t = \sigma(w^*_{rt}(h_{t-1}, x) + b)$

It has a candidate state that helps ~~where~~ in deciding that to update or reset to the previous value.

$\hat{h}_t = \tanh(w_{\hat{h}t} * z_t + w_{hx} r_t (h_{t-1}) + b)$

Now, this is faster method than LSTM as it has lesser no. of parameters & gates in the cell to get the values. Mathematically, it has 4 equations (GRU) while LSTM has 6 equations to work with. Now as the GRU has lesser parameters & equations, it will be easier to generalize and get output easily.

For example: If we need to translate 1 sentence from other language, instead of LSTM, GRU would be a better option to choose as, it will be faster easier to generalize & get the value.

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